CHAP3. IMAGE TRANSFORMS

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INTRODUCTION OF CHAP3

- Development of 2-D transforms and their properties
- 3.1 introduction the Fourier transform of one and two continuous variables
- 3.2 expressions in Discrete form
- 3.3 developing and illustrating several important properties of the 2-D F.T
- 3.4 F.F.T algorithm
- 3.5 Walsh,Hadamard,discrete cosine,Haar,and Slant transforms
- 3.6 introduction of Hotelling transforms and its applications

INTRODUCTION TO THE FOURIER TRANSFORM

f(x) : continuous function of a real variable

1-D Fourier transform pair

$$\Im[f(x)] = F(u) = \int_{-\infty}^{\infty} f(x)\exp[-j2\pi ux]dx$$

$$\Im^{-1}[F(u)] = f(x) = \int_{-\infty}^{\infty} F(u)\exp[j2\pi ux]du$$

f(x) ;continuous and integrable F(u) ; integrable

$$F(u) = R(u) + jI(u) = |F(u)|e^{j\Phi(u)}$$

$$|F(u)| = [R^2(u) + I^2(u)]^{1/2}$$ Fourier spectrum of f(x)

$$\Phi(u) = \tan^{-1}\left[\frac{I(u)}{R(u)}\right]$$ spectral density

$$P(u) = |F(u)|^2 = R^2(u) + I^2(u)$$ power spectrum of f(x)

Fig 3.1

INTRODUCTION TO THE FOURIER TRANSFORMS

2-D Fourier transform pair

$$\Im\{f(x,y)\} = F(u,v) = \int\int_{-\infty}^{\infty} f(x,y)\exp[-j2\pi(ux+vy)]dxdy$$

$$\Im^{-1}\{F(u,v)\} = f(x,y) = \int\int_{-\infty}^{\infty} F(u,v)\exp[j2\pi(ux+vy)]dudv$$

$$|F(u,v)| = [R^2(u) + I^2(u)]^{1/2}$$

$$\phi(u,v) = \tan^{-1}\left[\frac{I(u,v)}{R(u,v)}\right]$$

$$P(u,v) = |F(u,v)|^2 = R^2(u) + I^2(u)$$

Fig 3.2
Fig 3.3

THE DISCRETE FOURIER TRANSFORMS

$$\{f(x_0), f(x_0+\Delta x), f(x_0+2\Delta x), \ldots, f(x_0+[N-1]\Delta x)\}$$

discretized sequence of continuous function f(x)

Fig 3.4

Discrete Fourier transform pair

$$F(u) = \frac{1}{N}\sum_{x=0}^{N-1} f(x)\exp[-j2pux/N] \quad u=0,1,2,\ldots\ldots\ldots,N-1$$

$$f(x) = \sum_{u=0}^{N-1} F(u)\exp[j2pux/N] \quad x=0,1,2,\ldots\ldots\ldots,N-1$$

$$\Delta u = \frac{1}{N\Delta x}$$

SOME PROPERTIES OF THE TWO-DIMENSIONAL FOURIER TRANSFORM

$$D(u,v) = c\log[1+|F(u,v)|]$$

→ Fourier spectra range is much higher than display device so upper compensates

Fig 3.6

THE DISCRETE TRANSFORM

2- variable case DFT

$$F(u,v) = \frac{1}{MN}\sum_{x=0}^{N-1}\sum_{y=0}^{N-1} f(x,y)\exp[-j2p(ux/M+vy/N)] \quad \begin{matrix} u=0,1,2,\ldots\ldots\ldots,M-1 \\ v=0,1,2,\ldots\ldots\ldots,N-1 \end{matrix}$$

$$f(x,y) = \sum_{u=0}^{N-1}\sum_{v=0}^{N-1} F(u,v)\exp[j2p(ux/M+vy/N)] \quad \begin{matrix} x=0,1,2,\ldots\ldots\ldots,M-1 \\ y=0,1,2,\ldots\ldots\ldots,N-1 \end{matrix}$$

$$\Delta u = \frac{1}{M\Delta x}, \quad \Delta v = \frac{1}{N\Delta y}$$

M=N 일때

$$F(u,v) = \frac{1}{N}\sum_{x=0}^{N-1}\sum_{y=0}^{N-1} f(x,y)\exp[-j2p(ux+vy)/N] \quad \begin{matrix} u=0,1,2,\ldots\ldots\ldots,M-1 \\ v=0,1,2,\ldots\ldots\ldots,N-1 \end{matrix}$$

$$f(x,y) = \frac{1}{N}\sum_{u=0}^{N-1}\sum_{v=0}^{N-1} F(u,v)\exp[j2p(ux+vy)/N] \quad \begin{matrix} x=0,1,2,\ldots\ldots\ldots,M-1 \\ y=0,1,2,\ldots\ldots\ldots,N-1 \end{matrix}$$

SOME PROPERTIES OF THE TWO-DIMENSIONAL FOURIER TRANSFORM

3.3.1 Separability ; computation of the 2-D F.T as a series of 1-D transform

$$F(u,v) = \frac{1}{N}\sum_{x=0}^{N-1}\exp[-j2pux/N]\sum_{y=0}^{N-1} f(x,y)\exp[-j2pvy/N] \quad u,v=0,1,2.....N-1$$

$$f(x,y) = \sum_{u=0}^{N-1}\exp[j2pux/N]\sum_{v=0}^{N-1} F(u,v)\exp[j2pvy/N] \quad x,y=0,1,2.....N-1$$

$$F(u,v) = \frac{1}{N}\sum_{x=0}^{N-1} F(x,v)\exp[-j2pux/N]$$

$$f(x,v) = N\left[\frac{1}{N}\sum_{y=0}^{N-1} f(x,y)\exp[-j2pvy/N]\right]$$

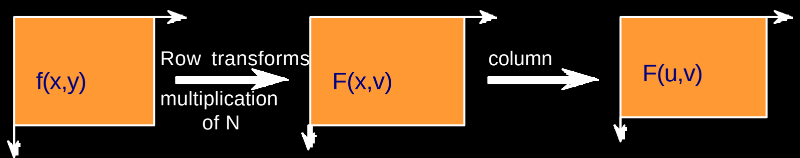


Fig 3.7 computation of 2-D F.T as a series of 1-D

SOME PROPERTIES OF THE TWO-DIMENSIONAL FOURIER TRANSFORM

3.3.2 Translation

$$f(x,y)\exp[j2\pi(u_0x+v_0y)/n] \Leftrightarrow F(u-u_0,v-v_0)$$

$$f(x-x_0,y-y_0) \Leftrightarrow F(u,v)\exp[-j2\pi(ux_0+vy_0)/N]$$

$$u_0=v_0=N/2 \qquad \exp[j2\pi(u_0x+v_0y)/N]=\exp[j\pi(x+y)]=(-1)^{(x+y)}$$

$$f(x,y)(-1)^{(x+y)} \Leftrightarrow F(u-N/2,v-N/2)$$

→ The origin of the F.T of f(x,y) -----> center of N*N frequency square

3.3.3 Periodicity and Conjugate symmetry

$$F(u,v)=F(u+N,v)=F(u,v+N)=F(u+N,v+N)$$

$$F(u,v)=\frac{1}{N}\sum_{x=0}^{N-1}\sum_{y=0}^{N-1} f(x,y)\exp[-j2\pi(ux+vy)/N]$$

$$F(u,v)=F^*(-u,-v) \qquad |F(u,v)|=|F(-u,-v)|$$ f(x,y) is real

Fig 3.8 Fig 3.9

SOME PROPERTIES OF THE TWO-DIMENSIONAL FOURIER TRANSFORM

3.3.6 Average Value

$$\bar{f}(x,y)=\frac{1}{N^2}\sum_{x=0}^{N-1}\sum_{y=0}^{N-1} f(x,y)$$

$$F(0,0)=\frac{1}{N}\sum_{x=0}^{N-1}\sum_{y=0}^{N-1} f(x,y)$$

$$\bar{f}(x,y)=\frac{1}{N}F(0,0)$$

3.3.7 Laplacian

$$\nabla^2 f(x,y)=\frac{\partial^2 f}{\partial x^2}+\frac{\partial^2 f}{\partial y^2}$$

$$\Im\{\nabla^2 f(x,y)\} \Leftrightarrow -(2\pi)^2(u^2+v^2)F(u,v)$$

→ useful for outlining edges in an image

SOME PROPERTIES OF THE TWO-DIMENSIONAL FOURIER TRANSFORM

3.3.4 Rotation ; Polar coordinates

$$x=r\cos\theta \quad y=r\sin\theta \quad u=w\cos\phi \quad v=w\sin\phi$$

$$f(x,y)\to f(r,\theta) \qquad F(u,v)\to F(w,\phi)$$

$$f(r,\theta+\theta_0)\to F(w,\phi+\theta_0)$$

Fig 3.10

3.3.5 Distributivity and scaling

$$\Im\{f_1(x,y)+f_2(x,y)\}=\Im\{f_1(x,y)\}+\Im\{f_2(x,y)\}$$

$$\Im\{f_1(x,y)*f_2(x,y)\}\neq\Im\{f_1(x,y)\}*\Im\{f_2(x,y)\}$$

$$af(x,y)\Leftrightarrow aF(u,v)$$

$$f(ax,by)\Leftrightarrow\frac{1}{|ab|}F(u/a,v/b)$$

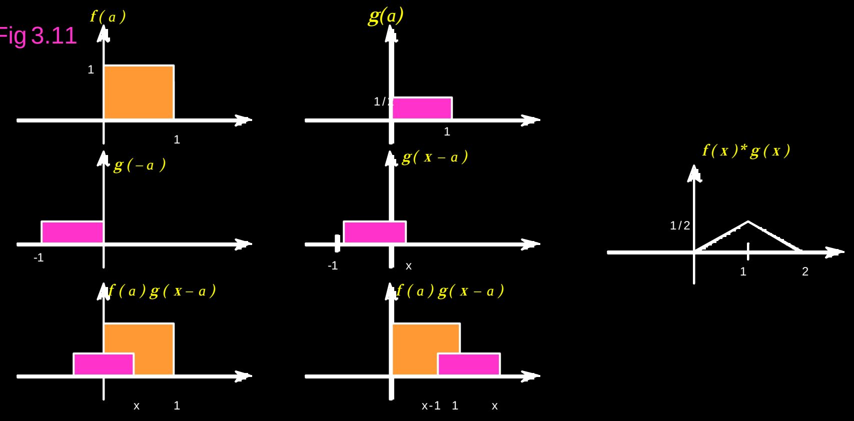
SOME PROPERTIES OF THE TWO-DIMENSIONAL FOURIER TRANSFORM

3.3.8 Convolution and Correlation ;

basic link between the spatial and frequency domains

Convolution : $f(x)*g(x)=\int_{-\infty}^{\infty} f(\alpha)g(x-\alpha)d\alpha$

Fig 3.11



SOME PROPERTIES OF THE TWO-DIMENSIONAL FOURIER TRANSFORM

Convolution Theorem $f(x)*g(x) \Leftrightarrow F(u)G(u)$

$$f(x)g(x) \Leftrightarrow F(u)*G(u)$$

f(x) and g(x) are discretized into sampled arrarys of A and B ,respectively.
For discrete convolution both functions have period M(M>=A+B-1)

extened sequences

$$f_e(x) = \begin{cases} f(x) & 0<=x<=A-1 \\ 0 & A<=x<=M-1 \end{cases}$$

$$g_e(x) = \begin{cases} g(x) & 0<=x<=B-1 \\ 0 & B<=x<=M-1 \end{cases}$$

$$f_e(x)*g_e(x) = \frac{1}{M}\sum_{m=0}^{M-1} f_e(m)g_e(x-m)$$

SOME PROPERTIES OF THE TWO-DIMENSIONAL FOURIER TRANSFORM

2-D Convolution

$$f(x,y)*g(x,y) = \int\int_{-\infty}^{\infty} f(a,b)g(x-a,y-b)dadb$$

$$f(x,y)*g(x,y) \Leftrightarrow F(u,v)G(u,v)$$

$$f(x,y)g(x,y) \Leftrightarrow F(u,v)*G(u,v)$$

to avoid wraparound error M>=A+C-1 x direction

N>=B+D-1 y direction

$$f_e(x,y) = \begin{cases} f(x,y) & 0<=x<=A-1 \text{ and } 0<=y<=B-1 \\ 0 & A<=x<=M-1 \text{ and } B<=y<=N-1 \end{cases}$$

$$g_e(x,y) = \begin{cases} g(x,y) & 0<=x<=C-1 \text{ and } 0<=y<=D-1 \\ 0 & C<=x<=M-1 \text{ and } D<=y<=M-1 \end{cases}$$

$$f_e(x,y)*g_e(x,y) = \frac{1}{MN}\sum_{m=0}^{N-1}\sum_{n=0}^{M-1} f_e(m,n)g_e(x-m,y-n)$$

SOME PROPERTIES OF THE TWO-DIMENSIONAL FOURIER TRANSFORM

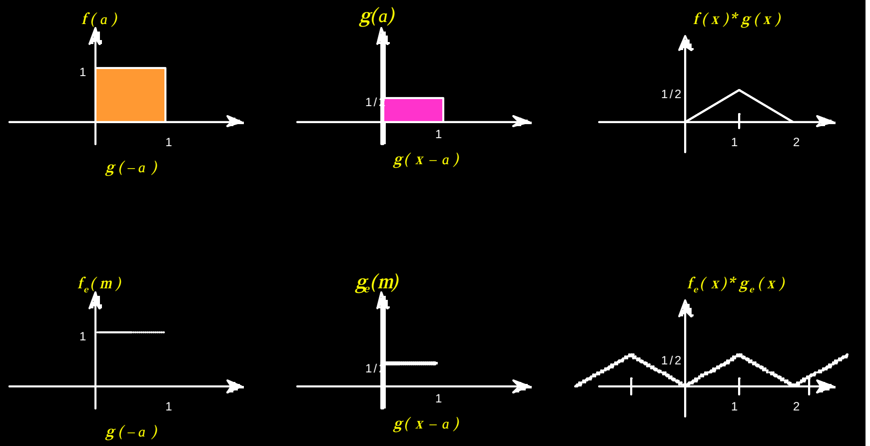


Fig 3.14

SOME PROPERTIES OF THE TWO-DIMENSIONAL FOURIER TRANSFORM

Correlation : $f(x)\circ g(x) = \int_{-\infty}^{\infty} f^*(a)g(x+a)da$

$$f_e(x)\circ g_e(x) = \frac{1}{M}\sum_{M=0}^{N-1} f^*_e(m)g_e(x+m)$$

x=0,1,2.....M-1

Fig 3.16

2-D Correlation

$$f(x,y)\circ g(x,y) = \int\int_{-\infty}^{\infty} f^*(a,b)g(x+a,y+b)dadb$$

$$f_e(x,y)\circ g_e(x,y) = \frac{1}{MN}\sum_{m=0}^{N-1}\sum_{n=0}^{M-1} f_e^*(m,n)g_e(x+m,y+n)$$

$$f(x,y)\circ g(x,y) \Leftrightarrow F^*(u,v)G(u,v)$$

$$f(x,y)^* g(x,y) \Leftrightarrow F(u,v)\circ G(u,v)$$

→ template,prototype matching

SOME PROPERTIES OF THE TWO-DIMENSIONAL FOURIER TRANSFORM

3.3.9 Sampling

1-D functions : band limited(W)

$$\Delta x <= \frac{1}{2W}$$ Whittaker-Shannonsamplingtheorem (Fig 3.17)

실제의 sampling은 finitesampling 이므로 windowing을 한다. (Fig 3.18)

$$h(x) = \begin{cases} 1 & 0 <= x <= X \\ 0 & elsewhere. \end{cases}$$ → frequencydomain에서 distortion이 일어남

$$\Delta u = \frac{1}{N\Delta x}$$ ← D.F.T (Fig 3.19)

THE FAST FOURIER TRANSFORM

$$F(u) = \frac{1}{N}\sum_{x=0}^{N-1} f(x)\exp[-j2pux/N]$$

→ number of complex multiplications and additions are proportional to N*N

3.4.1 FFT Algorithm

$$F(u) = \frac{1}{N}\sum_{x=0}^{N-1} f(x)W_N^{ux} \quad (W_N = \exp[-j2p/N])$$

$$N = 2^n = 2M \quad (M, N : Positive \quad integer)$$

$$F(u) = \frac{1}{2M}\sum_{x=0}^{2M-1} f(x)W_{2M}^{ux}$$

$$= \frac{1}{2}\left[\frac{1}{M}\sum_{x=0}^{M-1} f(2x)W_{2M}^{u(2x)} + \frac{1}{M}\sum_{x=0}^{M-1} f(2x+1)W_{2M}^{u(2x+1)}\right]$$

$$= \frac{1}{2}\left[\underbrace{\frac{1}{M}\sum_{x=0}^{M-1} f(2x)W_M^{ux}}_{F_{even}(u)} + \underbrace{\frac{1}{M}\sum_{x=0}^{M-1} f(2x+1)W_M^{ux}W_{2M}^{u(2x+1)}}_{F_{odd}(u)}\right] \leftarrow (W_{2M}^{u(2x)} = W_M^{ux})$$

u=0,1,2,.....M-1

SOME PROPERTIES OF THE TWO-DIMENSIONAL FOURIER TRANSFORM

2-D functions

$$\iint_{-\infty}^{\infty} f(x,y)d(x-x_0, y-y_0)dxdy = f(x_0, y_0)$$

$s(x,y)f(x,y)$ s(x,y) impulse train

$$\frac{1}{\Delta x} > 2W_u \qquad \frac{1}{\Delta y} > 2W_v$$

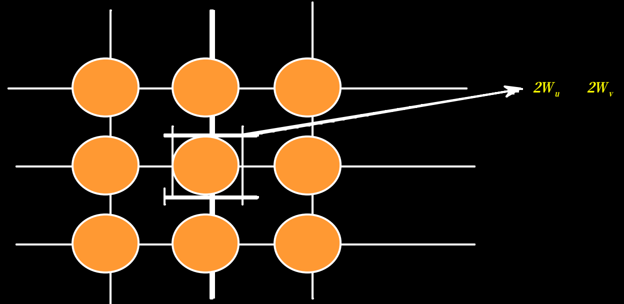


Fig 3.21 Frequency domain representation of a sampled 2-D, band-limited function

THE FAST FOURIER TRANSFORM

$$F(u) = \frac{1}{2}[F_{even}(u) + F_{odd}(u)W_{2M}^u]$$

$$F(u+M) = \frac{1}{2}[F_{even}(u) - F_{odd}(u)W_{2M}^u]$$

→ successivedoublingFFTalgorithm 사용

general number of multiplicaations and additions required to implement FFT

→ $m(n) = 2m(n-1) + 2^{n-1} \quad n >= 1$

$a(n) = 2a(n-1) + 2^n \quad n >= 1$

3.4.2 Number of operations

by induction, $m(n) = \frac{1}{2}2^n \log_2 2^n = \frac{1}{2}Nn \quad n >= 1$

$a(n) = 2^n \log_2 2^n = Nn \quad n >= 1$

$\rightarrow m(n+1) = 2m(n) + 2^n = 2(\frac{1}{2}Nn) + 2^n$

$= \frac{1}{2}(2^{n+1})(n+1)$

$\rightarrow a(n+1) = 2a(n) + 2^{n+1} = 2Nn + 2^{n+1}$

$= 2^{n+1}(n+1)$

THE FAST FOURIER TRANSFORM

3.4.3 The inverse FFT

1-D

$$F(u) = \frac{1}{N}\sum_{x=0}^{N-1} f(x)\exp[-j2pux/N]$$

$$f(x) = \sum_{u=0}^{N-1} F(u)\exp[j2pux/N]$$

$$\frac{1}{N}f^{*}(x) = \frac{1}{N}\sum_{u=0}^{N-1} F^{*}(u)\exp[-j2pux/N]$$

→ conjugate를 취하고 N을 곱하면 FFT algorithm으로 Inverse FFT를 구할수 있다.

2-D

$$F(u,v) = \frac{1}{N}\sum_{x=0}^{N-1}\sum_{y=0}^{N-1} f(x,y)\exp[-j2p(ux+vy)/N]$$

$$f(x,y) = \frac{1}{N}\sum_{u=0}^{N-1}\sum_{v=0}^{N-1} F(u,v)\exp[j2p(ux+vy)/N]$$

$$f^{*}(x,y) = \frac{1}{N}\sum_{u=0}^{N-1}\sum_{v=0}^{N-1} F^{*}(u,v)\exp[-j2p(ux+vy)/N]$$

THE FAST FOURIER TRANSFORM

3.4.4 Implementation ; a computer implementation of the FFT algorithm

In successive doubling methodfirst it needs bit-reversal rule for ordering input data

→ 110 --011 001--100 (table 3.2)

Fig 3.22

"DO 3" LOOP reorder the input data

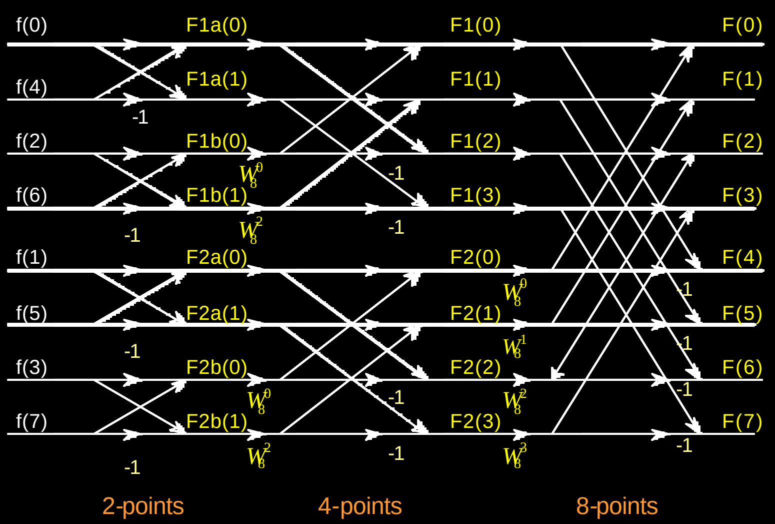
"DO 5" LOOP successive doubling calculation

"DO 6" LOOP divides the result by N

Fig 3.23

```
   SUBROUTINE FFT(F,LN)
   COMPLEX  F(1024),U,W,T,CMPLX
   PI=3.141593
   N=2**LN
   NV2=N/2
   NM1=N-1
   J=1
   D0  3  I=1,NM1
       IF(I.GE.J) GO TO 1
       T=F(J)
       F(J)=F(I)
       F(I)=T
1      K=NV2
2      IF(K.GE.J) GO TO 3
       J=J-K
       K=K/2
       GO  TO 2
3  J=J+K
   D0  5  L=1,LN
      LE=2**L
      LE1=LE/2
      U=(1.0,0.0)
      W=CMPLX(COS(PI/LE1),-SIN(PI/LE1))
      DO  5  J=1,LE11
          DO 4 I=J,N,LE
              IP=I+LE1
             T=F(IP)*U
             F(IP)=F(I)-T
4           F(I)=F(I)+T
5   U=U*w
    DO 6 I=1,N
6   F(I)=F(I)/FLOAT(N)
    RETURN
    END
```

THE FAST FOURIER TRANSFORM



OTHER SEPARABLE IMAGE TRANSFORMS

general transform and inverse transform

$$T(u) = \sum_{x=0}^{N-1} f(x)\underline{g(x,u)} \qquad u=0,1,2.......,N-1$$

forward transformation kernel

$$f(x) = \sum_{u=0}^{N-1} T(u)\underline{h(x,u)} \qquad x=0,1,2.......,N-1$$

inverse transformation kernel

→ kernel의 특성이 transformation의 특징에 영향을 줌

$$T(u,v) = \sum_{x=0}^{N-1} f(x,y)g(x,y,u,v)$$

$$f(x,y) = \sum_{u=0}^{N-1} T(u,v)h(x,y,u,v)$$

$$g(x,y,u,v) = g_1(x,u)g_2(y,v) \qquad ;\ separable$$

$$= g_1(x,u)g_1(y,v) \qquad ;\ symmetric$$

OTHER SEPARABLE IMAGE TRANSFORMS

A transform with a separable kernel can be computed in two steps

$$T(x,v)=\sum_{y=0}^{N-1} f(x,y)\,g_2(y,v) \qquad x,v=0,1,2,\ldots\ldots N-1$$

$$T(u,v)=\sum_{x=0}^{N-1} f(x,v)\,g_1(x,u) \qquad u,v=0,1,2,\ldots\ldots N-1$$

if the kernel g(x,y,u,v) is separable and symmetric

$$T = AFA$$

F ; N*N image matrix
A ; N*N symmetric transformation matrix with elements $a_{ij}=g_1(I,j)$
T ; the resulting N*N transformation for u,v

$$BTB = BAFAB$$

$$F = BTB \qquad (B = A^{-1})$$

$$\text{if} \quad B \neq A^{-1} \qquad \hat{F} = BAFAB$$ approximation of F

→ Fourier, Walsh, Hadamard, Discrete cosine, Haar, Slant transformss

OTHER SEPARABLE IMAGE TRANSFORMS

Inverse kernel and inverse Walsh transform

$$h(x,u)=\prod_{i=0}^{n-1}(-1)^{b_i(x)b_{n-1-i}(u)}$$

$$f(u)=\sum_{u=0}^{N-1} W(x)\prod_{i=0}^{n-1}(-1)^{b_i(x)b_{n-1-i}(u)}$$

2-D Walsh Transform

$$g(x,y,u,v)=h(x,y,u,v)=\frac{1}{M}\prod_{i=0}^{n-1}(-1)^{[b_i(x)b_{n-1-i}(u)+b_i(y)b_{n-1-i}(v)]}$$

$$W(u,v)=\frac{1}{N}\sum_{x=0}^{N-1}\sum_{y=0}^{N-1} f(x,y)\prod_{i=0}^{n-1}(-1)^{[b_i(x)b_{n-1-i}(u)+b_i(y)b_{n-1-i}(v)]}$$

$$f(x,y)=\frac{1}{N}\sum_{u=0}^{N-1}\sum_{v=0}^{N-1} H(u,v)\prod_{i=0}^{n-1}(-1)^{[b_i(x)b_{n-1-i}(u)+b_i(y)b_{n-1-i}(v)]}$$

OTHER SEPARABLE IMAGE TRANSFORMS

3.5.1 Walsh transform

$$g(x,u)=\frac{1}{M}\prod_{i=0}^{n-1}(-1)^{b_i(x)b_{n-1-i}(u)}$$

$$W(u)=\frac{1}{M}\sum_{x=0}^{N-1} f(x)\prod_{i=0}^{n-1}(-1)^{b_i(x)b_{n-1-i}(u)}$$

b_k is the kth bit in the binary representation of z (z=110=6 $b_0(z)=0, b_1(z)=1, b_2(z)=1$)

table 3.3 Value of the 1-D Walsh Transformation Kernel for N=8

u \ x	0	1	2	3	4	5	6	7
0	+	+	+	+	+	+	+	+
1	+	+	+	+	-	-	-	-
2	+	+	-	-	+	+	-	-
3	+	+	-	-	-	-	+	+
4	+	-	+	-	+	-	+	-
5	+	-	+	-	-	+	-	+
6	+	-	-	+	+	-	-	+
7	+	-	-	+	-	+	+	-

OTHER SEPARABLE IMAGE TRANSFORMS

$$g(x,y,u,v)=g_1(x,u)\,g_2(y,v)$$

$$=\left[\frac{1}{\sqrt{N}}\prod_{i=0}^{n-1}(-1)^{b_i(x)b_{n-1-i}(u)}\right]\left[\frac{1}{\sqrt{N}}\prod_{i=0}^{n-1}(-1)^{b_i(y)b_{n-1-i}(v)}\right]$$

→ separable, symmetric

FWT algorithm ; in FFT algorithm, $W_N=1$

$$W(u)=\frac{1}{2}[W_{even}(u)+W_{odd}(u)]$$

$$W(u+M)=\frac{1}{2}[W_{even}(u)-W_{odd}(u)]$$

Fig 3.25 Walsh basis function blocks for N=4

OTHER SEPARABLE IMAGE TRANSFORMS

3.5.2 Hadamard Transform

1-D Hadamard kernel and transform

$$g(x,u) = \frac{1}{N}(-1)^{\sum_{i=0}^{n-1} b_i(x)\, b_i(u)}$$

$$H(u) = \frac{1}{N}\sum_{x=0}^{N-1} f(x)(-1)^{\sum_{i=0}^{n-1} b_i(x)\, b_i(u)} \qquad u=0,1,2,\ldots\ldots N-1$$

$$f(x) = \sum_{u=0}^{N-1} H(u)(-1)^{\sum_{i=0}^{n-1} b_i(x)\, b_i(u)} \qquad x=0,1,2,\ldots\ldots N-1$$

2-D Hadamard

$$g(x,y,u,v) = h(x,y,u,v) = \frac{1}{N}(-1)^{\sum_{i=0}^{n-1}[b_i(x)b_i(u)+b_i(y)b_i(v)]}$$

$$H(u,v) = \frac{1}{N}\sum_{x=0}^{N-1}\sum_{y=0}^{N-1} f(x,y)(-1)^{\sum_{i=0}^{n-1}[b_i(x)b_i(u)+b_i(y)b_i(v)]}$$

$$f(x,y) = \frac{1}{N}\sum_{u=0}^{N-1}\sum_{v=0}^{N-1} H(u,v)(-1)^{\sum_{i=0}^{n-1}[b_i(x)b_i(u)+b_i(y)b_i(v)]}$$

OTHER SEPARABLE IMAGE TRANSFORMS

Walsh-Hadamard transform ← intermixed in image transform

successivedoublingformat

-> computation of FWT by a straightforward modfication of FFT

to obtain FHT algorithm ,it needs reordering the result of FWT

$$H_{2N} = \begin{bmatrix} H_N & H_N \\ H_N & -H_N \end{bmatrix} \qquad \text{Hadamard matrix}$$

$$A = \frac{1}{\sqrt{N}} H_N$$

sequency : the number of sign changes along a column of the Hadamrd matirx

→ in Table 3.4, sequencies 0,7,3,4,1,6,2, and 5

OTHER SEPARABLE IMAGE TRANSFORMS

$$g(x,y,u,v) = g_1(x,u)\,g_2(y,v) = h_1(x,u)\,h_2(y,v)$$

$$= \left[\frac{1}{\sqrt{N}}(-1)^{\sum_{i=0}^{n-1} b_i(x)b_i(u)}\right]\left[\frac{1}{\sqrt{N}}(-1)^{\sum_{i=0}^{n-1} b_i(y)b_i(v)}\right]$$

→ separable,symmetric

table 3.4 Value of the 1-D Hadamard Transformation Kernel for N=8

u \ x	0	1	2	3	4	5	6	7
0	+	+	+	+	+	+	+	+
1	+	-	+	-	+	-	+	-
2	+	+	-	-	+	+	-	-
3	+	-	-	+	+	-	-	+
4	+	+	+	+	-	-	-	-
5	+	-	+	-	-	+	-	+
6	+	+	-	-	-	-	+	+
7	+	-	-	+	-	+	+	-

OTHER SEPARABLE IMAGE TRANSFORMS

table 3.5 Value of the 1-D Ordered Hadamard Kernel for N=8

u \ x	0	1	2	3	4	5	6	7
0	+	+	+	+	+	+	+	+
1	+	+	+	+	-	-	-	-
2	+	+	-	-	-	-	+	+
3	+	+	-	-	+	+	-	-
4	+	-	-	+	+	-	-	+
5	+	-	-	+	-	+	+	-
6	+	-	+	-	-	-	-	+
7	+	-	+	-	+	+	+	-

→ sequencyincreasesasincreasingofu

OTHER SEPARABLE IMAGE TRANSFORMS

$$g(x,u) = \frac{1}{N}(-1)^{\sum_{i=0}^{n-1} b_i(x)p_i(u)}$$

$$p_0(u) = b_{n-1}(u), \quad p_1(u) = b_{n-1}(u) + b_{n-2}(u)$$

$$p_2(u) = b_{n-2}(u) + b_{n-3}(u)$$

$$\vdots$$

$$P_{n-1}(u) = b_1(u) + b_0(u)$$

1-D Hadamard kernel and transform (reordered)

$$g(x,u) = \frac{1}{N}(-1)^{\sum_{i=0}^{n-1} b_i(x)p_i(u)}$$

$$H(u) = \frac{1}{N}\sum_{x=0}^{N-1} f(x)(-1)^{\sum_{i=0}^{n-1} b_i(x)p_i(u)} \qquad u=0,1,2,......N-1$$

$$f(x) = \sum_{u=0}^{N-1} H(u)(-1)^{\sum_{i=0}^{n-1} b_i(x)p_i(u)} \qquad x=0,1,2,......N-1$$

OTHER SEPARABLE IMAGE TRANSFORMS

2-D DCT

$$C(u,v) = \alpha(u)\alpha(v)\sum_{x=0}^{N-1}\sum_{y=0}^{N-1} f(x,y)\cos\left[\frac{(2x+1)u\pi}{2N}\right]\cos\left[\frac{(2y+1)v\pi}{2N}\right]$$

$$f(x,y) = \sum_{u=0}^{N-1}\sum_{v=0}^{N-1} \alpha(u)\alpha(v)C(u,v)\cos\left[\frac{(2x+1)u\pi}{2N}\right]\cos\left[\frac{(2y+1)v\pi}{2N}\right]$$

Fig 3.28

3.5.4 The Haar Transform

Haar functions $h_k(z)$

→ continuous over [0,1] for k=0,1,2,....N-1

$k = 2^p + q - 1$ 0<=p<=n-1,q=0 or 1 for p=0,and 1<=q<=2^p for p !=0

$$h_0(z) = h_{00}(z) = \frac{1}{\sqrt{N}} \quad \text{for} \quad z \in [0,1]$$

$$h_k(z) = h_{pq}(z) = \frac{1}{\sqrt{N}}\begin{cases} 2^{p/2} & \frac{q-1}{2^p} <= z < \frac{q-1/2}{2^p} \\ -2^{p/2} & \frac{q-1/2}{2^p} <= z < \frac{q}{2^p} \\ 0 & \end{cases}$$

OTHER SEPARABLE IMAGE TRANSFORMS

2-D Hadamard (reordered)

$$g(x,y,u,v) = h(x,y,u,v) = \frac{1}{N}(-1)^{\sum_{i=0}^{n-1}[b_i(x)p_i(u)+b_i(y)p_i(v)]}$$

$$H(u,v) = \frac{1}{N}\sum_{x=0}^{N-1}\sum_{y=0}^{N-1} f(x,y)(-1)^{\sum_{i=0}^{n-1}[b_i(x)p_i(u)+b_i(y)p_i(v)]}$$

$$f(x,y) = \frac{1}{N}\sum_{u=0}^{N-1}\sum_{v=0}^{N-1} H(u,v)(-1)^{\sum_{i=0}^{n-1}[b_i(x)p_i(u)+b_i(y)p_i(v)]}$$

Fig 3.26

3.5.3 Discrete Cosine Transform

$$C(u) = a(u)\sum_{x=0}^{N-1} f(x)\cos\left[\frac{(2x+1)u\pi}{2N}\right] \qquad u=0,1,2,......N-1$$

$$f(x) = \sum_{u=0}^{N-1} a(u)C(u)\cos\left[\frac{(2x+1)u\pi}{2N}\right] \qquad x=0,1,2,......N-1$$

$$a(u) = \begin{cases} \sqrt{\frac{1}{N}} & u=0 \\ \sqrt{\frac{2}{N}} & u=1,2,......N-1 \end{cases}$$

OTHER SEPARABLE IMAGE TRANSFORMS

3.5.5 The Slant transform

$$S_N = \frac{1}{\sqrt{2}}\left[\begin{array}{cc|c|cc|c} 1 & 0 & & 1 & 0 & \\ a_N & b_N & 0 & -a_N & b_N & 0 \\ \hline \multicolumn{2}{c|}{0} & I_{(N/2)-2} & \multicolumn{2}{c|}{0} & I_{(N/2)-2} \\ \hline 0 & 1 & & 0 & -1 & \\ -b_N & a_N & 0 & b_N & a_N & 0 \\ \hline \multicolumn{2}{c|}{0} & I_{(N/2)-2} & \multicolumn{2}{c|}{0} & -I_{(N/2)-2} \end{array}\right]\left[\begin{array}{c|c} S_{N/2} & 0 \\ \hline 0 & S_{N/2} \end{array}\right]$$

$$S_2 = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$a_N = \left[\frac{3N^2}{4(N^2-1)}\right]^{1/2} \qquad b_N = \left[\frac{N^2-4}{4(N^2-1)}\right]^{1/2}$$

THE HOTELLING TRANSFORM

- Based on statistical properties of vector representations
- eigenvector,principal component , or discrete Karhunen-Loeve transform.

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ : \\ x_n \end{bmatrix}$$ pupulation of random vector

$$\mathbf{m}_x = E\{\mathbf{x}\}$$ meanvector

$$C_x = E\{(\mathbf{x} - \mathbf{m}_x)(\mathbf{x} - \mathbf{m}_x)^T\}$$ covariance matrix

→ element Cii is the vairance, Cij is the covariance

→ if Cij=Cji=0 , Xi,Xj are uncorrelated

$$\mathbf{m}_x = \frac{1}{M}\sum_{k=1}^{M} \mathbf{x}_k$$

$$C_x = \frac{1}{M}\sum_{k=1}^{M} \mathbf{x}_k \mathbf{x}_k^T - \mathbf{m}_x \mathbf{m}_x^T$$

THE HOTELLING TRANSFORM

reconstruction fo x from y

$$\mathbf{x} = A^T \mathbf{y} + \mathbf{m}_x$$

$$\hat{\mathbf{x}} = A_k^T \mathbf{y} + \mathbf{m}_x$$ K eigenvectors corresponding to the K largest eigenvalues (A is K*n)

$$e_{ms} = \sum_{j=1}^{n} l_j - \sum_{j=1}^{k} l_j = \sum_{j=k+1}^{n} l_j$$

Example

THE HOTELLING TRANSFORM

Cx is real and symmetric

→ finding a set of n orthonormal eigenvectors is possible

Let $\mathbf{e}_i$ and $l_i, i = 1,2,\ldots\ldots n$ be the eigenvectors and eigenvalues of Cx,

A be a matrix whose rows are formed from the eigenvectors of Cx

$l_j >= l_{j+1}$ (*for convenience*)

Hotelling transform $\mathbf{y} = A(\mathbf{x} - \mathbf{m}_x)$

$$\mathbf{m}_y = 0$$

$$C_y = A C_x A^T = \begin{bmatrix} l_1 & & & 0 \\ & l_2 & & \\ & & . & \\ 0 & & & l_n \end{bmatrix}$$

→ Cx and Cy have the same eigenvalues, and eigenvectors

Example

CONCLUDING REMARKS

- Theoretical foundation of image transforms and their properties
- Fourier transform -> FFT
- separability, centralization, convolution of the F.T
- Transform theory plays important role in image processing as a formal discipline